Soohyun Choi

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
Hanyang University • Information and Intelligence Systems Lab (IISL) • Seoul, Republic of Korea
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Research Interests

My research focuses on reinforcement learning, particularly the mathematical and geometric analysis of learning algorithms. I develop algorithms and modeling frameworks using generative models and stochastic processes for goal-conditioned and long-horizon control.

Education

Hanyang University Sep. 2024–Present
Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) • Advisor: Prof. Songnam Hong

Hanyang University 2018–2024
B.S. in Electronic Engineering

Preprints

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning 2026
Soohyun Choi, Seonvin Cho*, and Songnam Hong • *Equal contribution • arXiv:2608.29061 • Code*
Developed a hierarchical offline GCRL framework that connects subgoal selection to short-horizon execution through state-space bridges, inverse dynamics, and execution-time replanning.

Is One Step Enough for Offline Policy Improvement? 2026
Soohyun Choi, Seonvin Cho*, and Songnam Hong† • *Equal contribution • †Corresponding author • arXiv:2609.03842 v2 • Code*
Developed MPI, a re-centered proximal policy-improvement framework; distinguished fixed-total-horizon subdivision from additional refinement and studied geometry, discretization, and critic error with TD3+BC and Gaussian IQL extraction.

Role-Adaptive Policy Optimization for Offline Reinforcement Learning 2026
Seonvin Cho, Soohyun Choi, and Songnam Hong • arXiv:2609.40149
Developed RAPO to adapt policy-update coefficients through role-specific meta-gradients: separate execution and bootstrap actors in TD3+BC, and adaptive advantage-weighted policy extraction in IQL.

Ongoing Research

Extensions of PathBridger Present
Skill Learning and Composition: Investigating how to learn reusable skills from offline trajectories and compose them for long-horizon goal-conditioned tasks.
Latent-Space Bridge Planning: Exploring learned representations and geometrically structured latent spaces for goal-conditioned bridge generation and short-horizon execution.

Research Experience

Graduate Researcher, Information and Intelligence Systems Lab Sep. 2024–Present
Hanyang University • Advisor: Prof. Songnam Hong
Research on mathematical and geometric analysis for offline policy optimization and generative modeling for long-horizon control. Projects include MPI, RAPO, and PathBridger.

Research Participant, SK hynix–Hanyang University Project 2024–2025
REACT: Recursive Algorithm for RC Network Transfer Functions
Developed a recursive closed-form transfer-function method for arbitrary RC trees and a dominant-pole reduced-order approximation validated against SPICE.

Selected Research Project

ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold 2024; rev. 2026
Soohyun Choi, Minseok Oh, and Ickhyun Song • Undergraduate capstone technical report • Report
Formulated designer-selected performance equivalence classes and quotient geometry for circuit search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Honors & Funding

Top Excellence Award - Alumni Association Special Award 2023
Top Excellence Award - 15th Capstone Design Fair 2023
Graduation Honors, Hanyang University 2024
BK Research Assistantship, Hanyang University Fall 2025

Selected Graduate Coursework

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I